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ECOPATH II — a software for balancing steady-state ecosystem models and calculating network characteristics *

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ABSTRACT

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The ECOPATH II microcomputer software is presented as an approach for balancing ecosystem models. It includes (i) routines for balancing the flow in a steady-state ecosystem from estimation of a missing parameter for all groups in the system, (ii) routines for estimating network flow indices, and (iii) miscellaneous routines for deriving additional indices such as food selection indices and omnivory indices. The use of ECOPATH II is exemplified through presentation of a model of the Schlei Fjord ecosystem (Western Baltic).

INTRODUCTION

Since the International Biological Program (IBP) emphasized *ecosystem* research more than two decades ago, ecologists have studied what may be hundreds of systems or parts of systems worldwide. Thanks to the IBP the focus of many studies has been on describing flows in the systems and we now have well developed methodologies for measuring trophic interaction between most groups in a system (e.g., Vollenweider, 1969; Edmondson and Winberg, 1971; Holme and McIntyre, 1971; Bagenal, 1978; Fasham, 1984).

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While the IBP focused mainly on the lower part of the ecosystem, where the bulk of the flow occurs, developments in the 1980s have led to an improved picture of what is happening at the higher trophic levels of aquatic systems, especially of those that are commercially exploited. Notable here are a number of complex simulation models developed by fisheries biologists (e.g., Andersen and Ursin, 1977), some of which are now on the verge of serving as management tools (e.g., Sparre, 1991).

The ecosystem analyses of the IBP and follow-up studies have led to a large number of excellent scientific papers describing parts of ecosystems. It appears, however, that few of these studies have resulted in the presentation of balanced models of whole systems. We think this is due to the absence of a suitable tool, i.e., a versatile approach for balancing ecosystem models. Here we describe a program, the ECOPATH II software system, which may provide such an approach.

MODEL DESCRIPTION

Programming language

ECOPATH II is presently programmed in Microsoft Basic 7.0, Professional Developers Version, and is available with documentation from the authors in an executable version requiring no commercial software (Christensen and Pauly, 1991). It can be run on any IBM-compatible microcomputer. The present description relates to Version 2.0 of April 1991.

The architecture of ECOPATH II

The ECOPATH II model is developed from the ECOPATH model of Polovina (1984), with which it shares its "basic equation" (see below). This equation was originally proposed for the estimation of biomasses in steady-state ecosystems. Pauly et al. (1987) conceived ECOPATH II as consisting mainly of two interacting elements: (i) routines for estimating biomasses, or production/biomass ratios, as well as food consumption by the various elements (boxes) of a steady-state trophic model; and (ii) routines based on the theory of Ulanowicz (1986) for analyzing the flows estimated by applying (i) to data.

The version of ECOPATH II described here presents, in addition, a set of miscellaneous routines for deriving further statistics from the biomasses and flows estimated in (i), and further developing the theory in (ii). Notably, it incorporates an attempt to quantify a number of Odum's (1969) 24 indices of system maturity.

ECOPATH. The basic equations

It is assumed that the system to be modelled is in steady state. For each of the living groups in the system this implies that input equals output, i.e.

$$Q = P + R + U \tag{1}$$

where Q is consumption, P production, R respiration, and U unassimilated food. From this equation, the respiration can be estimated once the other flows have been quantified.

The production part of equation (1) is modelled explicitly in ECOPATH models. Basically, the approach is to model an ecosystem using a set of simultaneous linear equations (one for each group i in the system), i.e.

Production by (i) – all predation on (i) – non-predation losses of (i) – export of $(i) = 0$, for all i. This can be expressed as

$$P_i - M2_i - P_i(1 - EE_i) - EX_i = 0 \tag{2}$$

where P_i is the production of (i), $M2_i$ is the predation mortality of (i), EE_i is the Ecotrophic Efficiency of (i), $(1 - EE_i)$ is the "other mortality", and EX_i is the Export of (i).

Equation (2) can be re-expressed as

$$B_i \cdot PB_i - \sum_{j=1}^{n} B_j \cdot QB_j \cdot DC_{ji} - PB_i \cdot B_i \cdot (1 - EE_i) - EX_i = 0$$

or

$$B_i \cdot PB_i \cdot EE_i - \sum_{j=1}^{n} B_j \cdot QB_j \cdot DC_{ji} - EX_i = 0 \tag{3}$$

where B_i is the biomass of i, PB_i is the production/biomass ratio, QB_i is the consumption/biomass ratio and DC_{ji} is the fraction of prey (i) in the average diet of predator j.

Based on (3), for a system with n groups, n linear equations can be given; in explicit terms,

$$B_1PB_1EE_1 - B_1QB_1DC_{11} - B_2QB_2DC_{21} - \ldots - B_nQB_nDC_{n1} - EX_1 = 0 \tag{3.1}$$
$$B_2PB_2EE_2 - B_1QB_1DC_{12} - B_2QB_2DC_{22} - \ldots - B_nQB_nDC_{n2} - EX_2 = 0 \tag{3.2}$$
$$\vdots$$
$$B_nPB_nEE_n - B_1QB_1DC_{1n} - B_2QB_2DC_{2n} - \ldots - B_nQB_nDC_{nn} - EX_n = 0 \tag{3.n}$$

This system of simultaneous linear equations can be solved using standard matrix algebra.

If, however, the determinant of a matrix is zero, or if the matrix is not square, it has no ordinary inverse. Still, a generalized inverse can be found

in most cases (Mackay, 1981). In the ECOPATH II model, we have adopted the program of Mackay (1981) to estimate the generalized inverse.

If the set of equations (3.1)–(3.n) is overdetermined (more equations than unknowns), and the equations are not mutually consistent the generalized inverse method provides least squares estimates, which minimize the discrepancies.

Ancillary variables

To give guidance for the balancing of ecosystems a number of physiological variables characterizing groups has been included in ECOPATH II, e.g. gross and food conversion efficiencies, respiration/assimilation ratio, production/respiration ratio, and respiration/biomass ratio.

The requirements

The steady-state requirement of ECOPATH II may appear problematic, but should be taken as implying that the model outputs only apply to the period for which the inputs are deemed valid; the same requirements are implied when any rate variable is estimated for any mathematical representation of reality. For a fast-changing ecosystem such as an aquaculture pond, the steady-state assumption may perhaps be used for a model

TABLE 1

Input data for the Schlei Fjord ecosystem (based on Nauen, 1984). Units: t/km^2; rates are yearly. Dashes show parameters subsequently calculated by ECOPATH II [a]

Group	Catch	Biomass	Production biomass	Consumption biomass
1. Apex predator	0.03	0.1	0.7	6.7
2. Med. predator	0.38	–	1.0	11.3
3. Planktivores	0.05	–	1.8	17.1
4. Temp. planktiv.	2.30	–	1.7	16.4
5. Whitefish	0.12	–	0.5	10.8
6. Small fish	0.00	–	0.2	9.1
7. Zoobenthos	0.00	96.9	1.4	–
8. Zooplankton	0.00	10.2	10.0	36.5
9. Phytoplankton	0.00	89.2	–	n.a.
10. Detritus	0.00	100.0	n.a.	n.a.

[a] The following additional assumptions were made: (i) other mortality (flow to detritus0 = 5% of production (Groups 2–6 and 9; calculated for Groups 1, 7 and 8); (ii) gross food conversion efficiency of zoobenthos = 0.15.

TABLE 2

Food consumption matrix for the Schlei Fjord system, expressed in percent of consumption on a weight basis. Dashes indicate no consumption (modified from Nauen, 1984)

Prey	Predator							
	1	2	3	4	5	6	7	8
1. Apex predator	–	–	–	–	–	–	–	–
2. Med. predator	20	–	–	–	–	–	–	–
3. Planktivores	50	–	–	–	–	–	–	–
4. Temp. planktiv.	–	–	–	–	–	–	–	–
5. Whitefish	5	2.6	–	–	–	–	–	–
6. Small fish	25	48.7	8.3	–	–	–	–	–
7. Zoobenthos	–	48.7	–	–	100	80	–	–
8. Zooplankton	–	–	91.7	100	–	20	–	–
9. Phytoplankton	–	–	–	–	–	–	5	80
10. Detritus [a]	–	–	–	–	–	–	95	20

[a] Nauen (1984) does not consider detritiory; our interpretation of trophic flows in Schlei Fjord suggests detritiory to be likely for both zoobenthos (95%) and zooplankton (20%).

describing 1 month, while for a coral reef model a decade may be appropriate.

To illustrate the data requirements for ECOPATH II, we have given the input data for a model of the Schlei Fjord ecosystem (Western Baltic Sea) in Tables 1 and 2 (based on data in Nauen, 1984). The derived flow diagram for the Schlei Fjord system is shown in Fig. 1. To increase the descriptive and explanatory impact of the flow diagram, and to facilitate comparisons between ecosystems we are using some constructional rules. Note that (1) the boxes are placed on the y-axis according to trophic level of the groups, (2) the areas of the boxes are scaled after the logarithms of the group biomasses, (3) flows exiting a box do so from the upper half of the box, while flows entering a box do so via the lower half of the box, and (4) flows exiting a box cannot branch, but they can be linked with flows exiting other boxes. The flows are balanced so that input equals output for all boxes.

Ascendency

ECOPATH II links concepts developed by theoretical ecologists, especially the theory of Ulanowicz (1986), with those used by biologists working in fisheries and aquaculture. Most notable is the inclusion of a routine for calculating "ascendency" in the form suggested by Ulanowicz and Norden (1990). Ascendency is a measure of average mutual information in a system, derived from information theory and scaled by system throughput.

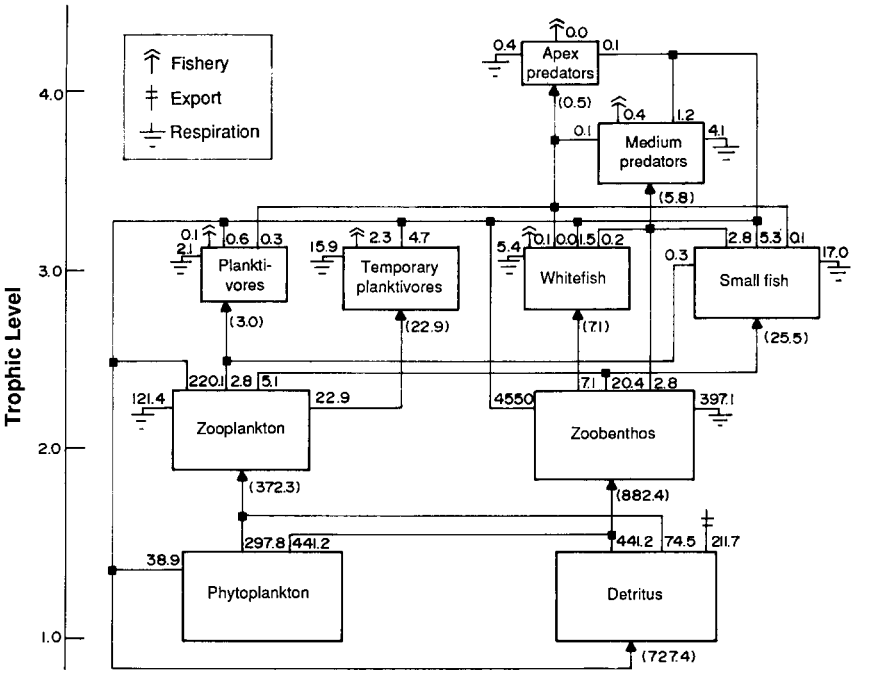


Fig. 1. Flow diagram of the Schlei Fjord system (based on Nauen, 1984). Flows are expressed in t/km^2/year.

Thus, if one knows the location of a unit of energy the uncertainty of where it will go next is reduced by an amount known as the "average mutual information"

$$I = \sum_{i=1}^{n} \sum_{j=1}^{n} f_{ij} \cdot Q_i \cdot \log\left(f_{ij} / \sum_{k=1}^{n} f_{kj} \cdot Q_k \right)$$

where n is the number of groups in the system, and if T_{ij} is a measure of the energy flow from j to i, f_{ij} is the fraction of the total flow from j that is represented by T_{ij}, or

$$f_{ij} = T_{ij} / \sum_{k=1}^{n} T_{kj}$$

Q_i is the probability that a unit of energy passes through i, or

$$Q_i = \sum_{k=1}^{n} T_{ki} / \sum_{l=1}^{n} \sum_{m=1}^{n} T_{lm}$$

Q_i, as a probability, is scaled by multiplication with the total throughput of the system, T, where

$$T = \sum_{i=1}^{n} \sum_{j=1}^{n} T_{ij}$$

Further

$$A = T \cdot I$$

where it is A that is called "ascendency". The ascendency is symmetrical and will have the same value whether calculated from input or output.

There is an upper limit for the size of the ascendency. This upper limit is called the "development capacity" and is estimated from

$$C = H \cdot T$$

where H is called the "statistical entropy", and is estimated from

$$H = -\sum_{i=1}^{n} Q_i \cdot \log Q_i$$

The difference between capacity and ascendency is called "system overhead". The overheads provide limits on how much the ascendency can increase and reflect the system's "strength in reserve" from which it can draw to meet unexpected perturbations (Ulanowicz, 1986)

Judged from the theoretical foundation of the concept it has been stated that ascendency "correlates well with most of Odum's (1969) 24 properties of 'mature' ecosystems" (Ulanowicz and Norden, 1990).

However, only a few ecosystems have been analyzed using Ulanowicz's theory. It thus remains to be shown how close the correlation is between ascendency and maturity.

Trophic level

Lindeman (1942) introduced the concept of trophic levels. Further, by treating the ecosystem as a thermodynamic unit, he could describe the efficiencies of transfers between trophic levels. However, some authors disagreed. Cousin (1985), noting that "a hawk feeds on five trophic levels", suggested abandoning the trophic level concept.

Alternatively, species can be placed on fractional trophic levels, as suggested by Odum and Heald (1975). ECOPATH II includes such fractional trophic levels.

Detritus and primary producers such as phytoplankton and benthic producers have, by definition, a trophic level equal to unity. For all other groups the (mean weighted) trophic level (TL) of group (i) is defined as

one plus the sum of the trophic levels of its preys multiplied by the prey's proportion in the diet of species (i), or,

$$TL = 1 + \sum_{j=1}^{n} DC_{ij} \cdot TL_j$$

where DC_{ij}, referred to as the diet composition, is the proportion of prey (j) in the diet of species (i), TL_j is the trophic level of prey (j), and n is the number of groups in the system.

The trophic levels of groups other than primary producer or detritus may be expressed as a system of equations in the form:

$$\begin{array}{llll} 1 = & TL_1(1-DC_{11}) & -TL_1DC_{12} & -TL_1DC_{13} \quad \ldots -TL_1DC_{1n} \\ 1 = & -TL_2DC_{21} & +TL_2(1-DC_{22}) & -TL_2DC_{23} \quad \ldots -TL_2DC_{2n} \\ \vdots & \vdots & \vdots & \vdots \quad \ldots \quad \vdots \\ 1 = & -TL_nDC_{n1} & -TL_nDC_{n2} & -TL_nDC_{n3} \quad \ldots +TL_n(1-DC_{nn}) \end{array}$$

This equation system is solved using a standard inverse method. Following this approach a consumer eating 40% plants ($TL = 1$) and 60% herbivores ($TL = 2$) will have a trophic level of $1 + [0.4 \cdot 1 + 0.6 \cdot 2] = 2.6$.

Trophic aggregation

In addition to the calculation of fractional trophic levels, we have included a routine that aggregates the entire system into discrete trophic levels sensu Lindeman. This routine is based on an approach suggested and described by Ulanowicz (in press); it reverses the routine for calculation of fractional trophic levels. Thus, 40% of the flow through the consumer

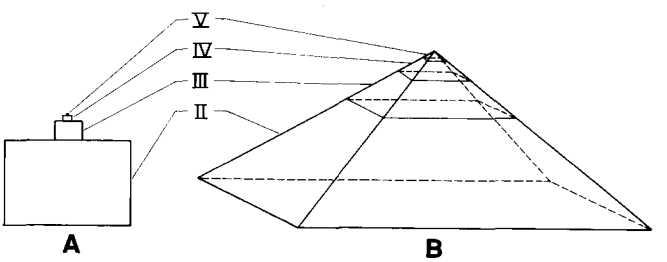


Fig. 2. Two representations of trophic pyramids of ecosystems (here: Schlei Fjord). (A) The traditional Lindeman pyramid, as depicted in various ecology textbooks (e.g., Krebs, 1972). Often, but not necessarily, the area of each trophic level is proportional to the biomass or throughput at that level. (B) The proposed new version of the trophic pyramid. The volume of each discrete trophic level (Roman numerals) is proportional to the throughput at that level.

group mentioned above would be attributed to the herbivore level and 60% to the first-order carnivory level.

This leads to a further development of the pyramid metaphor: one can give them three dimensions, just as they have in Egypt. An example of this is given for the Schlei Fjord system in Fig. 2, where the volume of each of the pyramidal compartments representing discrete trophic levels is proportional to the total throughput of the trophic level.

The trophic aggregation produces as its main result calculated estimates of trophic transfer efficiencies by trophic levels, e.g., the transfer efficiencies in the Schlei Fjord system are 4.9% for the herbivory level, and 10.3, 8.2 and 6.1% for the first three carnivory levels, respectively.

Omnivory index

Pauly et al. (1987) introduced the concept of 'omnivory index' to partly describe the feeding behavior of the consumer groups. The omnivory index (OI) is calculated as the variance of the trophic levels of a consumer's preys. For group (i) we have,

$$\mathrm{OI}_i = \sum_{j=1}^{n} (TL_j - TL)^2 \cdot DC_{ij}$$

where n is the number of groups in the system, TL_j is the trophic level of prey j, TL is the average trophic level of the preys, i.e. one less than the trophic level of predator i, and DC_{ij} is the fraction of prey (j) in the average diet of predator (i).

If a predator only has prey on one trophic level its omnivory index will equal zero, while a large omnivory index indicates that the trophic positions of a predator's preys are variable. Cousin's hawk would have a high omnivory index.

Selection indices

One of the most widely used indices for selection is the Ivlev electivity index, E_i (Ivlev, 1961) defined as

$$E_i = (r_i - P_i)/(r_i + P_i)$$

where r_i is the relative abundance of a prey in a predator's diet and P_i is the prey's relative abundance in the ecosystem. In ECOPATH II, the r_i and P_i refer to biomass, not numbers. E_i is scaled so that $E_i = -1$ corresponds to total avoidance, $E_i = 0$ represents non-selective feeding, and $E_i = 1$ shows exclusive feeding.

We have included the Ivlev electivity index since it is often used in the literature. This index has, however, a major shortcoming, seriously limiting its usefulness as a selection index. As shown by several authors, e.g. Jacobs (1974), the Ivlev index is not independent of prey density.

A better approach may be to use the standardized forage ratio (S_i) as suggested by Chesson (1983). This index, which is independent of prey availability, is given by

$$S_i = (r_i/P_i)/\left(\sum_{j=1}^{n} r_j/P_j\right)$$

where r_i and P_i are defined as above and n is the number of groups in the system.

As implemented in ECOPATH II, the forage ratio has been transformed such as to vary between -1 and 1, where -1, 0 and 1 can be interpreted as the Ivlev index.

Recycling index

An index of how much of the flow of an ecosystem is recycled has been included in ECOPATH II. This recycling index, developed by Finn (1976), is expressed as percentage of total throughput. It was originally intended to quantify one of Odum's (1969) properties of system maturity. However, its interpretation is apparently not as simple as E.P. Odum conceived, with recycling increasing as a system matures. Wulff and Ulanowicz (1989) suggest that the opposite may indeed be the case.

Cycles and pathways

A routine based on an approach suggested by Ulanowicz (1986) has been implemented to describe the numerous cycles and pathways that are implied in an ecosystem (Table 3).

In addition, a measure of the average path length is included, defined as the average number of groups/boxes a flow passes through. The average path length (PL) is calculated from a steady-state version of the equation presented by Finn (1976). We have

$$\mathrm{PL} = T/\left[\sum_{i=1}^{n} EX_i + \sum_{i=1}^{n} R_i\right]$$

where T is the total systems throughput, n is the number of groups in the system, EX_i is all exports from group (i), and R_i is the respiration of group (i).

TABLE 3

Example of an output from the "Cycles and Pathways" routine of ECOPATH II, showing all pathways leading from the primary producers (phytoplankton) to the apex predators in the Schlei Fjord ecosystem model in Fig. 1

1. Phytoplank.	→	Zoobenth.	→	Whitefish	→	Med. pred.	→	Apex pred.
2. Phytoplank.	→	Zoobenth.	→	Small fish	→	Med. pred.	→	Apex pred.
3. Phytoplank.	→	Zooplank.	→	Small fish	→	Med. pred.	→	Apex pred.
4. Phytoplank.	→	Zoobenth.	→	Med. pred.	→	Apex pred.	→	Apex pred.
5. Phytoplank.	→	Zoobenth.	→	Small fish	→	Planktiv.	→	Apex pred.
6. Phytoplank.	→	Zooplank.	→	Small fish	→	Planktiv.	→	Apex pred.
7. Phytoplank.	→	Zoobenth.	→	Planktiv.	→	Apex pred.		
8. Phytoplank.	→	Zoobenth.	→	Whitefish	→	Apex pred.		
9. Phytoplank.	→	Zoobenth.	→	Small fish	→	Apex pred.		
10. Phytoplank.	→	Zooplank.	→	Small fish	→	Apex pred.		

Total number of pathways = 10
Mean pathway length = 3.6

Mixed trophic impacts

Leontief (1951) developed a method to reveal the direct and indirect interactions in the economy of the USA, using what has since been called the Leontief matrix. This matrix was introduced to ecology by Hannon (1973) and Hannon and Joiris (1989). The latter developed the method further so that it becomes possible to give qualitative statements of the impact of direct and indirect interactions (including competition) in a system.

Ulanowicz and Puccia (1990) developed a similar approach, and a routine based on their method has been implemented in the ECOPATH II system. In this approach the positive effect (g_{ij}) a prey (j) has upon a predator (i) is expressed as the proportion the prey constitutes to the diet of the predator. We have

$$g_{ij} = DC_{ij}$$

The native impact (f_{ij}) a predator (i) has upon its prey (j) is expressed as the fraction of the total predation on the prey that is caused by the predator

$$f_{ij} = B_i \cdot QB_i \cdot DC_{ij} / \sum_{k=1}^{n} B_k \cdot QB_k \cdot DC_{kj}$$

where n is the number of groups in the system.

The net impact (q_{ij}) of j upon i is found as the differences between the positive and the negative impacts

$$q_{ij} = g_{ij} - f_{ji}$$

The q_{ij}'s in the system are components of an $n \times n$ matrix, which we will call Q. If it is assumed that the overall trophic impact of any pathway is found as the product of all the q's involved, and further that the combined effect of pathways joining is the sum of the individual pathways, it is possible to use the same matrix methodology as in standard economic input–output analysis.

Therefore, the total impact of one group on another can be estimated as

$$[M] = \sum_{h=1}^{\infty} [Q]^h \tag{4}$$

where $[M]$ is the total mixed trophic impact matrix. We know that the series in equation (4) will converge to

$$\sum_{h=0}^{\infty} [Q]^h = \{[I] - [Q]\}^{-1}$$

where $[I]$ is the identity matrix, and the exponent -1 indicates matrix inversion. As $[Q]^0 = [I]$ we have

$$[M] = \{[I] - [Q]\}^{-1} - [I]$$

so that the mixed trophic impact can readily be calculated using simple matrix manipulations including a standard inverse method.

An example of the use of mixed trophic impact is given in Fig. 3 for Schlei Fjord. To facilitate the interpretation of this figure note that the apex predators have a slight positive impact on the small fishes inspite of the occurrence of a direct predation. In this case the indirect effect of apex predation on the medium predators (which are the most important predators on small fishes) weighs more in the overall accounting.

In addition to showing the trophic impacts, this routine can be interpreted as a form of sensitivity analysis. In the Schlei Fjord example one can, for example, conclude that whitefish does not have a strong impact on any of the other groups, while zoobenthos is of major importance for several other groups.

Aggregation of boxes

The ascendency as well as certain other features of an ecosystem are affected by the number of groups that is included in the system description. Ulanowicz (1986) therefore suggested an algorithm for aggregation of

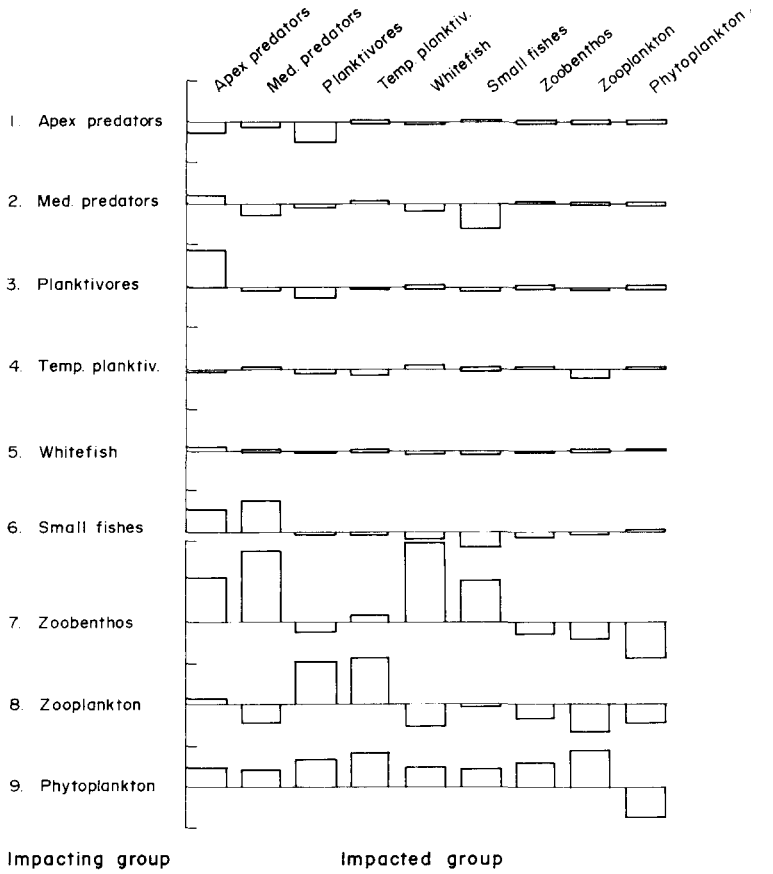


Fig. 3. Mixed trophic impacts in the Schlei Fjord ecosystem. The figure shows the direct and indirect impacts on the living groups in the system caused by the groups given at the left. Positive impacts are shown above the base line, negative below. The impacts are relative, but comparable between groups.

groups, based on stepwise combination of the pairs of groups that cause the least reduction in system ascendency. We have included a routine based on this suggestion in ECOPATH II. When used on the Schlei Fjord system, the results in Fig. 4 emerge. Noteworthy (from the inset) is that the ascendency diminishes notably only when the system is aggregated to less than six groups. Further, the resulting aggregations are closely related to the trophic levels of the groups. Thus the figure shows a remarkable resemblance to figures illustrating clustering techniques, but this is coincidental: trophic level is not an input parameter for the routine.

Both features of the aggregation, i.e. that the ascendency only drops off when a system is aggregated to very few groups, and that the groupings are

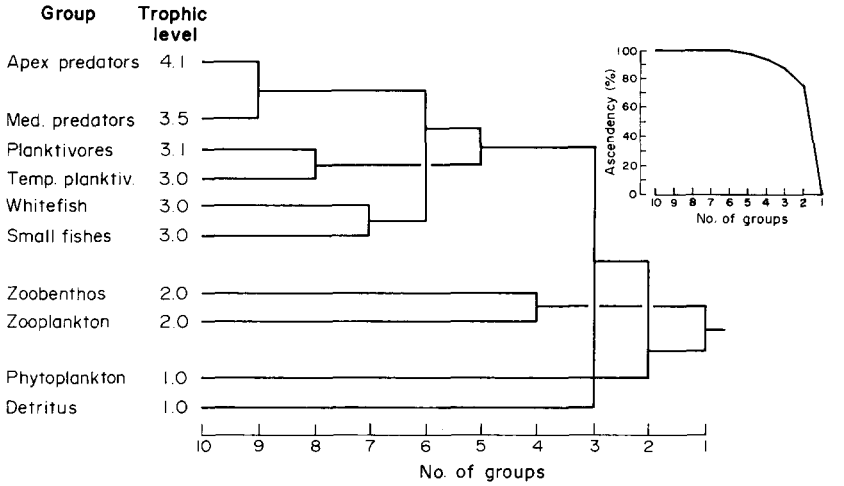


Fig. 4. The main graph (Schlei Fjord) shows the pairwise aggregation of groups that results in the last decrease in system ascendency. Note that groups with similar trophic levels are aggregated first. The small inset shows the decrease in ascendency resulting from the aggregation.

closely linked with trophic levels, are generally valid, as shown by analysis of more than 30 ecosystems (unpublished data). Our findings support those of Ulanowicz (1986) who used the routine on the Crystal River system and found that the aggregation gave the same result as "intuitive guesswork". At the same time, this feature of ascendency raises the question of whether it is indeed an appropriate measure of ecosystem "growth and development" as suggested by Ulanowicz (1986). Clearly the magnitude of the ascendency is primarily a function of the flow between the few large groups in the lower parts of the system. Ascendency therefore cannot capture the dynamics of the interaction in the upper parts of the systems. Similar considerations led H.T. Odum to suggest that ascendency should be calculated using "emergy" (embodied energy; Odum, 1988) related units instead of direct flows (R.E. Ulanowicz, personal communication, 1990).

Trophic level and mortality

Size and mortality are related. This has been shown by many authors, e.g. Pauly (1980), McGurk (1986), and Gulland (1987). Further, for groups in a steady-state system, the instantaneous rate of mortality is equal to the production/biomass ratio (P/B) (Allen, 1971). In ECOPATH II, we have linked trophic level (TL) with the inverse of the P/B ratio, i.e. the

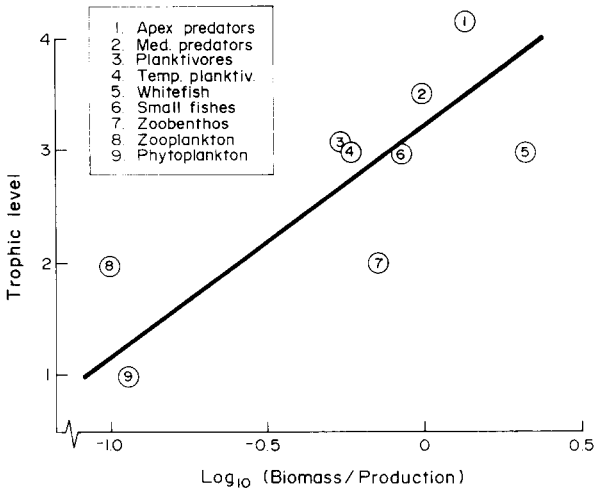


Fig. 5. TLBP plot showing correlation between trophic level and (log) biomass/production. The slope of the GM regression line is 2.07. This slope, we suggest, is a system-specific index characterizing the flow up the food web.

biomass/production (B/P) ratio, and found for all ecosystems so far studied a close positive correlation between these variables. This is illustrated in Fig. 5. This plot, which we have tentatively called a "TLBP plot", gives a characteristic picture of the ecosystem. We suggest use of the slope of the geometric mean regression of the TLBP plot as an index of the interaction of trophic level and mortality in an ecosystem.

CONCLUSION

The ECOPATH II system as briefly documented here is an attempt to present an approach and a software that should be useful to any fishery biologist or aquatic ecologist attempting to look at more than two species at a time. It has been used by various authors to describe over 30 different ecosystems, as diverse as, for example, rice-fish culture systems, Chinese carp polyculture ponds, estuaries, upwelling systems, coral reefs, shelves and the open oceans (Christensen and Pauly, in press).

We see the interest for this approach as an indication of the need for a model that would make the transition from readily available population characteristics to balanced ecosystem models a feasible and even easy step. We intend to carry the development of ECOPATH II further, and encourage researchers with interest in using or helping to develop further the system to contact us.

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